MIL-P-17098A
14 OCTOBER 1952
SUPERSEDING
MIL-P-17098(Aer)
1 September 1952

MILITARY SPECIFICATION

PSYCHROMETER, HAND, ELECTRIC, ASPIRATING; ML-450( )/UM AND CARRYING CASE, PSYCHROMETER, CY-1238( )/UM

This specification has been approved by the Departments of the Army, the Navy, and the Air Force.

1. SCOPE AND CLASSIFICATION

1.1 Scope.—This specification covers a portable, power driven psychrometer with carrying case used for determining dew point and relative humidity of the atmosphere.

1.2 Classification.—Psychrometers covered by this specification shall be of a type designated ML-450( )/UM.

2. APPLICABLE SPECIFICATIONS, OTHER PUBLICATIONS AND DRAWINGS

2.1 General.—The following specifications, publications and drawings of the issue in effect on the date of invitation for bids, form a part of this specification.

2.2 Specifications.

Military Specifications

MIL-B-131—Barrier Material: Water-Vapor Proof, Flexible.

MIL-P-6064—Packaging of light-weight aircraft accessories.

JAN-P-100—Packaging and Packing for Overseas Shipment. General Specification.

JAN-P-106—Packaging and Packing for Overseas Shipment — Boxes, wood nailed.

JAN-P-127—Packaging and Packing for Overseas Shipment — Tape, adhesive, pressure-sensitive, water-resistant.

2.3 Other Specifications.

Military Standards

MIL-STD-125—Manual of Standard Descriptions for Electronic Equipment and Material.

MIL-STD-129—Marking of Shipments.

National Bureau of Standards

Handbook H28—Screw Thread Standards for Federal Service.

2.4 Availability of Specifications, Publications, and Drawings.

2.4.1 When requesting specifications, publications and drawings refer to both title and number.

2.4.2 Copies of this specification and applicable specifications may be obtained upon application to the Commanding Officer, U. S. Naval Air Station, Johnsville, Pennsylvania, Attention: Technical Records Division.

3. REQUIREMENTS

3.1 Component Parts.—The psychrometers shall consist essentially of a case, two thermometer tubes, fan and motor, light and carrying case.

3.2 Preproduction Samples.—Preproduction samples will be required by this specification. They shall be submitted to the Government agency concerned.

3.2.1 *Samples.*—After award of contract, the contractor shall proceed with fabrication of and shall assemble samples of psychrometers, using tools and methods which are the same, or representative of those to be used in quantity produc-

227519—53

Source: https://assist.dla.mil -- Downloaded: 202610-18T09:45Z
Check the source to verify that this is the current version before use.

tion. Two samples shall be submitted for all tests of this specification and other tests not specifically described herein when such tests are deemed necessary to determine full compliance with the specifications.

3.2.1.1 *Test Data.*—Test data and engineering reports showing compliance with the specification shall accompany the preproduction samples.

3.2.2 *Sample Shipment.*—The samples shall be marked and shipped in accordance with directions given in the contract and specifications.

3.2.3 *Approval.*—Approval of the submitted samples shall not be construed as authorizing any deviations from the requirement of this specification. If any sample is disapproved, the contractor may be required to submit additional samples until the required quantity of satisfactory samples has been submitted. Such additional samples shall be accompanied by a description of the changes made to correct the faults of the rejected samples.

3.3 **Service Condition Requirement.**—The equipment shall be capable of satisfactory operation and storage under any possible combination of the service conditions listed below:

(a) Temperature—Any temperature in the range 0° F. to +120° F.

(b) Relative Humidity—Any relative humidity up to 100 percent including condensation caused by temperature changes.

(c) Fungi and Insects—As encountered in tropical service.

(d) Vibration and Shock—Vibration and shock as encountered aboard ship and in A/C transportation and as specified in paragraph 4.3.5.

(e) Salt-laden Air — As encountered aboard ship.

In addition the equipment shall be capable of meeting the tests in Section 4.

3.4 **Material.**

3.4.1 *Material.*—The material for each part shall be as specified herein. The best material for tropical and Arctic use commercially available for the purpose shall be used when a definite material is not designated. Wherever possible, metals used shall be non-corroding, if corrodible metals are used they shall, wherever possible, be adequately protected from corrosion by plating, lacquer, baked enamel, or other suitable covering or finish; finishes which will crack, chip or peel under normal usage or when subjected to changes of temperature shall not be employed. Wear surfaces of gear teeth, pivots, bearing faces, etc., which cannot be protected with finishes, shall be made of a material which will not appreciably tarnish or form oxidation films.

3.4.2 *Case.*—The case shall be of spun or fabricated construction.

3.4.3 *Wick.*—The wick shall be of muslin, or other suitable material, sewn or woven into a tube of such diameter as to fit snugly over the thermometer bulb.

3.4.4 *Intake Shield.*—The intake shields, within which the wet and dry bulb thermometers are set, shall be as thin as is consistent with sturdiness and ability to withstand hard usage and shall be of suitable material and finished with a reflective finish to minimize the influences of instrumental temperature upon the thermometer bulbs.

3.4.5 *Undesignated Parts.*—Undesignated parts shall be of non-corrosive materials. No two metals shall be used in contact if their characteristics are such that contact between them results in accelerated corrosion in one of the materials.

3.4.6 *Screws.*—All screws, nuts, and other similar parts used in the assembly of the instrument shall conform to the Handbook H28, and shall be Class 2 fit unless otherwise required by design. In general, screws in hard material shall be fine thread series. All screws, nuts, bolts, and washers shall be protected against corrosion.

3.4.6.1 *Set Screws.*—Set screws shall be of the hardened cup-pointed type.

3.4.7 *Critical Material.*—The contractor may substitute non-critical materials for critical materials during wartime upon receiving approval of the contracting agency.

3.4.8 *Tropicalization.*—The parts, sub-assemblies, and assembled equipment shall be treated for moisture and fungus resistance.

3.5 **Detailed Construction Requirements.**

3.5.1 *Wet and Dry Bulb Thermometers.*—Both thermometers shall be essentially identi-

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T09:45Z
Check the source to verify that this is the current version before use.

cal and shall be of the mercury-in-glass type, and graduated $4\frac{1}{2}'' \pm \frac{1}{4}''$ in length in whole degress Fahrenheit. The scale shall have a minimum range of 100° F. from plus 10° F. (±5° F.) to plus 110° F. The scale shall have each tenth graduation numbered. All graduations and numerals shall be clearly etched into the glass and filled with dark permanent pigment. The width of the graduations shall not be greater than two-tenths the distance between adjacent graduations. Numbering shall be adjacent to the mercury column, so located that the mercury column is seen between the ends of graduations and numerals. The size of the numerals shall be such as to be clearly readable at arm's length by an observer of normal vision. The scale and numerals shall be so positioned on the stem that all graduations and numerals will be clearly visible from a single point directly in front of the assembled psychrometer. The design of the bulb shall be such that the wick can be replaced easily. The thermometer shall not be adversely affected by temperature of +150° F. or −40° Fahrenheit.

3.5.2 *Thermometer Accuracy.*—The maximum allowable error in either thermometer shall be 0.5° Fahrenheit and matched pairs of thermometers supplied for the psychrometer shall be in agreement within 0.3° Fahrenheit.

3.5.3 *Motor and Fan.*—A motor-operated fan properly governed to provide the optimum essentially-constant air speed for proper ventilation of the thermometer bulbs shall be incorporated in the design. A positive means for starting and stopping the fan shall be provided. Means shall be provided so that the motor may be operated with or without the lights. The design shall be such that, in event of motor failure, the complete motor and fan assembly may be easily removed and replaced as a unit, without the use of special tools.

3.5.4 *Thermometer Holder or Frame.*—The design of the frame shall be such that while ventilation takes place, it shall be possible to observe the thermometer scales at all times without interruption in the operation of the instrument, so that the true maximum wet bulb depression shall not be missed.

3.5.5 *Illumination.*—The illumination shall be a variable low intensity red light. The light from the instrument shall provide for adequate protection of the "dark conditioning" of the operator's eyes and inconspicuous enough not to reveal the operator's position.

3.5.6 *Water Reservoir.*—The water reservoir shall be an integral part of the instrument, and shall be of suitable size. Means for manually moistening the wick of the wet bulb thermometer at intervals from the water reservoir, shall also be provided.

3.5.7 *Water Pump.*—The water pump shall be an integral part of the water reservoir. It shall operate freely and shall deliver directly the correct amount of water to wet the wet bulb wick.

3.5.8 *Neck Strap.*—A neck strap of grey woven nylon webbing shall be supplied with a convenient snap fastener for attaching to the psychrometer.

3.5.9 *Carrying Case.*—A metal carrying case properly padded to guard the instrument against shocks occasioned by normal usage or by being dropped shall be provided.

3.5.9.1 *Carrying Case Components.*—The carrying case shall provide for the storage of two (2) spare thermometers, a water bottle of four ounce capacity and necessary field replacement spare parts.

3.5.10 *Nameplate.*—A metal nameplate of a size consistent with legibility and space available for mounting shall be attached to a convenient location on the psychrometer housing and carrying case. The nameplate shall contain, in any suitable arrangement, the information pertinent to the type of gear as shown in the sample below:

BUREAU OF AERONAUTICS
PSYCHROMETER ML-450( )/UM
Stock No.....
Contract No.....
Serial No.....
Manufactured by (Manufacturer's name)

3.5.11 *Psychrometer Graph.*—A properly simplified, expanded psychrometer graph shall be suitably reproduced on the sides of the psychrometer.

3.5.12 *Spare Parts.*—Two spare thermometers, a water bottle of four ounce capacity, a light bulb, and necessary field replacement spare parts shall be provided.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T09:45Z
Check the source to verify that this is the current version before use.

4. SAMPLING, INSPECTION, AND TEST PROCEDURE

4.1 Classification of Tests.—The inspection and testing of the psychrometer shall be classified as follows:

4.1.1 *Preproduction Tests.*—Preproduction tests are those tests listed below which will be performed on samples submitted to determine if the samples comply with the requirements of this specification.

4.1.1.1 *Production Tests.*—All routine tests performed on production units to determine compliance with this specification for acceptance under contract.

4.1.2 *Inspection Procedures.*—Inspection procedures shall conform to and be performed in the order shown in Table I.

TABLE I.—*Inspection Procedure*

Inspection or Test	Paragraph Reference	Preproduction	Production Acceptance Inspection
Visual and Mechanical	4.2	x	x
Test at Room Temperature	4.3.2	x	x
Test at +40° F	4.3.3	x	
Temperature Change Test	4.3.4	x	
Shock Test	4.3.5	x	

All other tests necessary to determine conformance with this specification may be performed as follows:

(a) Non-destructive test on preproduction samples.

(b) Destructive tests on preproduction samples.

4.1.3 *Production Standard.*—Upon satisfactory completion of the tests on preproduction samples, such changes shall be made as are necessitated by the tests. One finally approved sample shall be retained in the possession of the contracting officer until the end of production. The other finally approved sample shall be returned to the contractor's plant to be used by the inspector as a standard of workmanship and performance. The production standard shall be currently modified in accordance with any and all subsequently approved changes.

4.1.4 *Facilities for Tests.*—All materials and processes entering into the fabrication shall be subject to the inspection of the authorized Government Inspector. The contractor shall furnish all necessary facilities and equipment for making the production tests and inspections required by this specification, and shall carry out all tests under the supervision of the inspector. The test equipment shall be adequate in quantity to enable the inspector to keep up with production.

4.2 Visual and Mechanical Inspection.—The equipment shall be given a visual and mechanical inspection to determine the quality of material, workmanship, and compliance with the requirements of this specification.

4.3 Test Procedure.

4.3.1 *Test of Psychrometer ML–450( )/UM.*—The contractor shall furnish necessary facilities and equipment for making all tests and inspection hereinafter required and shall carry out all tests under the supervision of the inspector. Unless otherwise specified in the contract, all instruments on the order shall be subjected to the tests and inspection required. The inspector shall ascertain that the psychrometer meets all the requirements of this specification. The Government reserves the right to make other tests not specifically described herein when such tests are deemed necessary to determine full compliance with the specification requirements. The contractor shall correct all deviations from this specification pointed out by the inspector.

4.3.2 *Comparison with Aspirating Psychrometer.*—Psychrometer ML–413( )/UM, completely assembled, shall be compared to a standard aspirating psychrometer of known accuracy. Calculated values of relative humidity shall be made by using U. S. Weather Bureau Psychrometric Tables, Bulletin No. 235. With room temperature 75° F. (±5° F.) and the average relative humidity in the room 40 percent (±10 percent), Psychrometer ML–450( )/UM shall be checked 3 times against the standard psychrometer. Both psychrometers shall be operated simultaneously, within one foot of each other, and at the same height above the floor. The time shall be noted when the wick is wetted and at the same instant the motor of the psychrometer is started. At the end of 2 minutes, wet and dry bulb thermometer readings shall be taken for both psychrometers. No calculated relative humidity (3 test readings) made from the Psychrometer ML–450( )/UM shall differ by more than 2

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T09:45Z
Check the source to verify that this is the current version before use.

percent from those calculated from the standard psychrometer readings taken at the same instant.

4.3.3 *Comparison with Aspirating Psychrometer in a Refrigerated Space.*—Preproduction model Psychrometer, completely assembled, shall be compared in a refrigerated space to a standard psychrometer of known accuracy. Calculated values of relative humidity shall be made by using U. S. Weather Bureau Psychrometric Tables, Bulletin No. 235. (The refrigerated space shall be of sufficient size to admit two operators to perform the test.) With the temperature of the refrigerated space 40° F. (±5° F.) and the average relative humidity in the space 50 percent (±10 percent), Psychrometer ML–450( )/UM shall be checked 3 times against the standard psychrometer. Both psychrometers shall be operated simultaneously within one foot of each other, and at the same height above the floor. The time the wick is wetted, at the same instant the motor of the psychrometer is started shall be noted by the use of a stopwatch. At the end of 2 minutes, wet and dry bulb thermometer readings shall be taken for both psychrometers and then the psychrometers stopped. No calculated relative humidity value (3 test readings) made from the psychrometer shall differ by more than 2 percent from those calculated from the standard psychrometer readings taken at the same instant.

4.3.4 *Temperature Change Test.*—Preproduction Model Psychrometer ML–450( )/UM, completely assembled, shall now be checked for humidity response with a temperature change. It shall be compared for each test reading to a standard psychrometer of known accuracy. Calculated values of relative humidity shall be made by using U. S. Weather Bureau Psychrometric Tables, Bulletin No. 235. With the room temperature outside the refrigerated space 75° F. (±5° F.) and the average relative humidity in the room outside the refrigerated space 50 percent (±10 percent), and the temperature of the refrigerated space 40° F. (±5° F.) and the average relative humidity in the space 50 percent (±10 percent), Psychrometer ML–450( )/UM shall be allowed to remain at rest in the room at 75° F. (±5° F.) for 30 minutes. It shall then be taken immediately into the refrigerated space, the wick properly wetted, the motor started, and the time noted. Both psychrometers shall be operated simultaneously, within one foot of each other, and at the same height above the floor. At the end of 2 minutes, wet and dry bulb thermometer readings shall be taken for both psychrometers and then the psychrometers stopped.

Psychrometer ML–450( )/UM shall then be allowed to remain at rest in the refrigerated space for 30 minutes. It shall then be taken immediately from the refrigerated space into the 75° F. (±5° F.) room, the wick properly wetted, the motor started, and the time noted. Both psychrometers shall be operated simultaneously, within one foot of each other and at the same height above the floor. At the end of 2 minutes, wet and dry bulb thermometer readings shall be taken for both psychrometers and then the psychrometers shall be stopped.

No calculated relative humidity value (2 test readings) made from the psychrometer shall differ by more than 4 percent from those calculated from the standard psychrometer readings taken at the same instant.

4.3.5 *Shock Test.*—This test will be performed by the Aerology Branch as part of the tests of the preproduction model. Only the preproduction model shall be subjected to this test. The packed instrument, as specified in paragraph 5.1, shall be tumbled end-over-side-over-end for four complete revolutions on a concrete surface. The box shall then be rotated 90° about its vertical axis and tumbled side-over-side-over-side for four complete revolutions onto a concrete surface. The packed instrument shall then be dropped on each of its eight corners from a height of 30 inches. The box shall then be opened and the instrument examined. There shall be no evidence of damage as a result of this test. All other tests of the preproduction model shall be made subsequent to this test.

5. PREPARATION FOR DELIVERY

5.1 **Packaging and Packing.**—The equipment shall be packaged and packed in accordance with Specification MIL–6064, Group III. The contract shall specify whether packing shall be domestic or for oversea shipment.

Source: https://assist.dla.mil -- Downloaded: 202640-18T09:45Z
Check the source to verify that this is the current version before use.

When overseas shipment is specified the equipment shall be packed in boxes in accordance with Specification JAN-P-106.

5.2 **Marking.**—All shipments shall be marked in accordance with MIL-Std-129.

6. NOTES

6.1 **Exhibits.**—Prospective suppliers may examine a Psychrometer ML-450( )/UM by making arrangements to do so with the Aerology Branch (MA-5) of the Bureau of Aeronautics. The sample psychrometer is representative of the general type of instrument desired.

6.2 **Notice.**—When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation or conveying any rights or permission to manufacture, use or sell any patented invention that may in any way be related thereto.

Custodians:
Army—Signal Corps
Navy—Bureau of Aeronautics

Other interest:
Army—MT
Navy—Md.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T09:45Z
Check the source to verify that this is the current version before use.

SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity (as indicated on reverse hereof).

SPECIFICATION

ORGANIZATION (of submitter) | CITY AND STATE

CONTRACT NO. | QUANTITY OF ITEMS PROCURED | DOLLAR AMOUNT
$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)* | DATE

Source: https://assist.dla.mil -- Downloaded: 20260-18T09:45Z
Check the source to verify that this is the current version before use.

FOLD

DEPARTMENT OF THE NAVY

POSTAGE AND FEES PAID
NAVY DEPARTMENT

OFFICIAL BUSINESS

COMMANDING GENERAL
U. S. ARMY ELECTRONICS COMMAND
ATTN: AMSEL-PP-E-SED
FORT MONMOUTH, N. J. 07703

FOLD

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T09:45Z
Check the source to verify that this is the current version before use.